

A matrix has zero as an eigenvalue if and only if:

Select one:

- ☒ a. It is singular
- ☐ b. It is orthogonal
- ☐ c. None of these
- ☐ d. It is invertible

Which statement is **true** for any $A \in \mathbb{R}^{m \times n}$?

- ☐ A and A^T have the same eigenvectors
- ☐ A and A^T have the same rank only if $m=n$
- ☒ A and A^T have the same singular values
- ☐ A and A^T have the same SVD

Consider the inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2$$

on $\mathbb{R}^2$. Which pair is orthogonal?

- ☐ a. $\mathbf{u} = (1, 2), \quad \mathbf{v} = (2, 1)$
- ☒ b. $\mathbf{u} = (1, 1), \quad \mathbf{v} = (1, -1)$
- ☐ c. $\mathbf{u} = (1, 0), \quad \mathbf{v} = (-3, 2)$
- ☐ d. $\mathbf{u} = (1, 0), \quad \mathbf{v} = (0, 1)$

Which property is satisfied by every inner product?

Select one:

- ☐ a. $\langle x+y, z \rangle = \langle x, z \rangle$
- ☐ b. $\langle x, y \rangle = -\langle y, x \rangle$
- ☒ c. $\langle x, y \rangle = \langle y, x \rangle$
- ☐ d. $\langle x, x \rangle < 0$

Let $S = \{(1, 1, 1), (1, 2, 3), (1, 3, 6)\}$. Which of the following is true?

Select one:

- ☐ a. S is linearly dependent because the third vector is a linear combination of the first two.
- ☒ b. S is a basis for $\mathbb{R}^3$ because the determinant of the matrix formed by the vectors is non-zero.
- ☐ c. S does not span $\mathbb{R}^3$ because it has only 3 vectors.
- ☐ d. S contains only two independent vectors because the third vector is a linear combination of the first two.

Given $u = (2, 7, 3, 1)$ and $v = (4, 1, -2, 5)$, which vector is orthogonal to both u and v ?

Select one:

- ☐ a. $(1, 0, 1, -2)$
- ☐ b. $(3, -1, 2, 0)$
- ☒ c. $(0, -1, 2, 1)$
- ☐ d. $(5, -2, 1, 1)$

Let $V = \mathbb{P}_4(\mathbb{R})$, and

$$W = \{p(x) \in V \mid p(0) = 0, p(1) = 0\}.$$

What is $\dim(W)$?

Select one:

- ☐ a. 2
- ☒ b. 3
- ☐ c. 4
- ☐ d. 5

Let $u=(k,1,2)$ and $v=(1,2,2)$. If the angle between u and v is 60°

Select one:

- ☐ a. $3(k+6)^2 = 25(k^2+5)$
- ☐ b. $4(k+6)^2 = 25(k^2+5)$
- ☒ c. $4(k+6)^2 = 9(k^2+5)$
- ☐ d. $9(k+6)^2 = 4(k^2+5)$

After Gaussian elimination, a system is represented by:

$$[1 \ 3 \ -2 \ | \ 1]$$

$$[0 \ 1 \ 4 \ | \ -2]$$

$$[0 \ 0 \ 0 \ | \ 0]$$

The number of variables is 3. What is true about the solution

Select one:

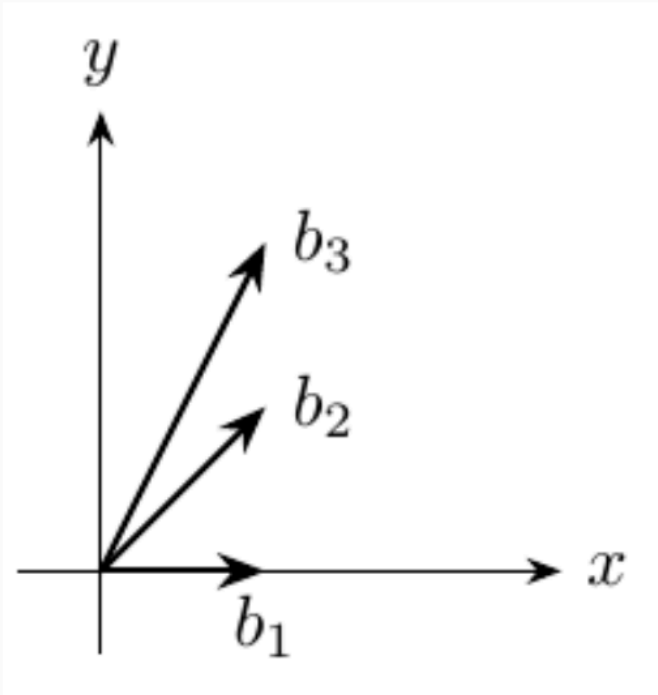
- ☐ a. No solution
- ☐ b. Unique solution
- ☐ c. Inconsistent system
- ☒ d. Infinitely many solutions

If V is an n -dimensional vector space, then the dimension of

Select one:

- ☐ a. n
- ☐ b. $2n$
- ☒ c. n^2
- ☐ d. $n!$

The vectors b_1, b_2, b_3 form a basis B of V . Which statement is true?



Select one:

- ☒ a. Every vector in V has exactly one representation as a linear combination of the vectors in B .
- ☐ b. Every vector in V has exactly two representations as a linear combination of the vectors in B .
- ☐ c. Some vectors in V have no representation using B .
- ☐ d. The representation is unique only if B is orthonormal.

On $\mathbb{R}^2$, define $\langle x, y \rangle = 2x_1y_1 + x_1y_2 + x_2y_1 + 2x_2y_2$.

Select one:

- ☐ A. 0
- ☐ B. 1
- ☐ C. 2
- ☒ D. $\sqrt{2}$

In Gaussian elimination, if you obtain a row of the form $[0 \ 0 \ 0 \ c]$

Select one:

- ☐ a. The system has a unique solution
- ☐ b. The system has infinitely many solutions
- ☐ c. More information is needed
- ☒ d. The system has no solution

Which of the following statements about Cholesky decompo

Select one:

- ☒ a. Every symmetric matrix has a Cholesky decomposition
- ☐ b. Every symmetric positive definite matrix has a Chole
- ☐ c. If $A=LL^T$ is a Cholesky decomposition, then L has p
- ☐ d. Every diagonal matrix with positive diagonal entries l

The **reduced row echelon form** of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ is

Select one:

- ☐ A. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix}$
- ☐ B. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$
- ☐ C. $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$
- ☒ D. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$

Let $V = \mathbb{P}_2(\mathbb{R})$, and
 $W = \{p(x) \in V \mid p(2) = 0\}$.

What is $\dim(W)$?

Select one:

- ☐ a. 3
- ☐ b. 1
- ☒ c. 2
- ☐ d. 0

The homogeneous system

$$A\mathbf{x}$$

has nontrivial solutions if

Select one:

- ☒ a. $\det(A) = 0$
- ☐ b. A is invertible
- ☐ c. $\det(A) \neq 0$
- ☐ d. $\text{rank}(A) = n$

A matrix A is Hermitian if:

Select one:

- ☐ a. $A^T = A$
- ☒ b. $A^H = A$
- ☐ c. $\det(A) = 1$
- ☐ d. A is lower triangular.

For which values of λ are the following 2×2 matrices linearly independent?

$$A_1 = \begin{bmatrix} 1 & \lambda \\ 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

- ☐ a. $\lambda = -2$
- ☐ b. $\lambda = 1$
- ☐ c. $\lambda = 2$
- ☒ d. $\lambda = 0$

Which of the following matrices are positive definite?

[I] $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

[II] $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

[III] $\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$

[IV] $\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$

☐ a. II and IV

☒ b. I and IV

☐ c. II and III

☐ d. I and III